# Day 21 | Thursday

21

December 3, 2026

## Checking square, diagonals and layout

12:45 PM–2:00 PM • Math • 75 min

## SOH CAH TOA and missing sides

2:15 PM–3:30 PM • Math • 75 min

**Week 6 • 2 scheduled blocks • 150 teaching minutes**

Estimate. Show the method. Label units. Check the result.

# Today’s exact schedule

## 12:45 PM–2:00 PM • 75 min

### Checking square, diagonals and layout

| | | |
|---|---|---|
| 12:45 PM–12:50 PM | **Start** | 5 |
| 12:50 PM–12:55 PM | **Recall** | 5 |
| 12:55 PM–1:10 PM | **Teach** | 15 |
| 1:10 PM–1:25 PM | **Together** | 15 |
| 1:25 PM–1:45 PM | **Practise** | 20 |
| 1:45 PM–1:55 PM | **Check** | 10 |
| 1:55 PM–2:00 PM | **Wrap up** | 5 |

Math • Math Week 6, Lesson 7

## 2:15 PM–3:30 PM • 75 min

### SOH CAH TOA and missing sides

| | | |
|---|---|---|
| 2:15 PM–2:20 PM | **Start** | 5 |
| 2:20 PM–2:25 PM | **Recall** | 5 |
| 2:25 PM–2:40 PM | **Teach** | 15 |
| 2:40 PM–2:55 PM | **Together** | 15 |
| 2:55 PM–3:15 PM | **Practise** | 20 |
| 3:15 PM–3:25 PM | **Check** | 10 |
| 3:25 PM–3:30 PM | **Wrap up** | 5 |

Math • Math Week 6, Lesson 8

**Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time**

# Checking square, diagonals and layout

**Today’s goal: Apply a scaled 3–4–5 triangle and compare diagonals to diagnose an out-of-square frame.**

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 12.5–12.6 pp.304–306

Exercise Book 12A Q17–18 (p.115); 12B Q4–5,7 (pp.116–117)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 12.5–12.6 pp.304–306

# Recall and reconnect

Discuss which three lengths make a scaled 3–4–5 triangle and why both diagonals matter.

**Attempt the retrieval task independently; explain the method before checking.**

Resources: Exercise Book 12A Q17–18 (p.115); 12B Q4–5,7 (pp.116–117)

Textbook sections 12.5–12.6 pp.304–306

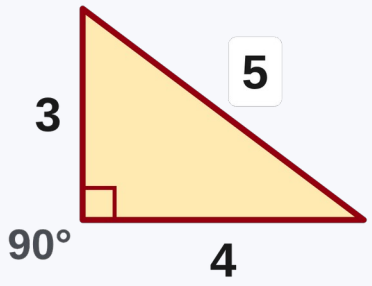
# Use a scaled 3–4–5 triangle

**1** $3^2 + 4^2 = 5^2$, so a 3–4–5 triangle has a right angle.

**2** Multiply all three sides by the same factor to scale the check.

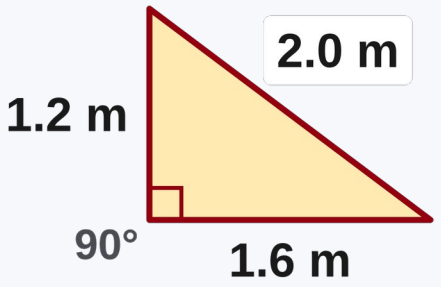
**3** Measure the diagonal between the two marked side lengths.

**4** Compare with the expected diagonal before deciding how to adjust the corner.




$3^2 + 4^2 = 5^2$

right angle



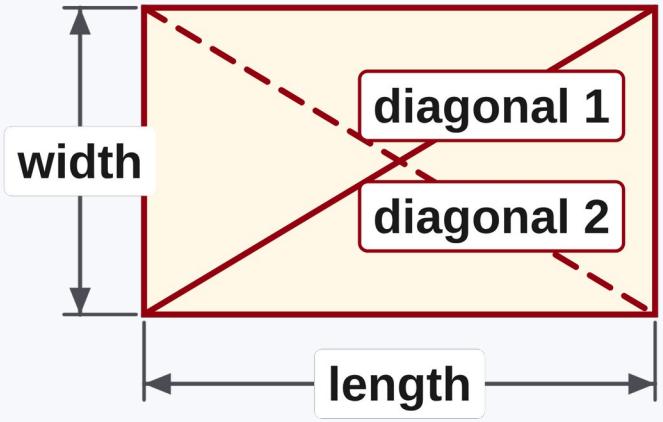

every side × 0.4

same right angle

Textbook sections 12.5–12.6 pp.304–306

# Check the complete frame

1. Calculate the expected diagonal from the stated side lengths.
2. Compare both measured diagonals with the target and with each other.
3. Use the book diagram to identify an open or closed corner.
4. Recheck dimensions and diagonals after the indicated correction.




**Target = √(length² + width²)**

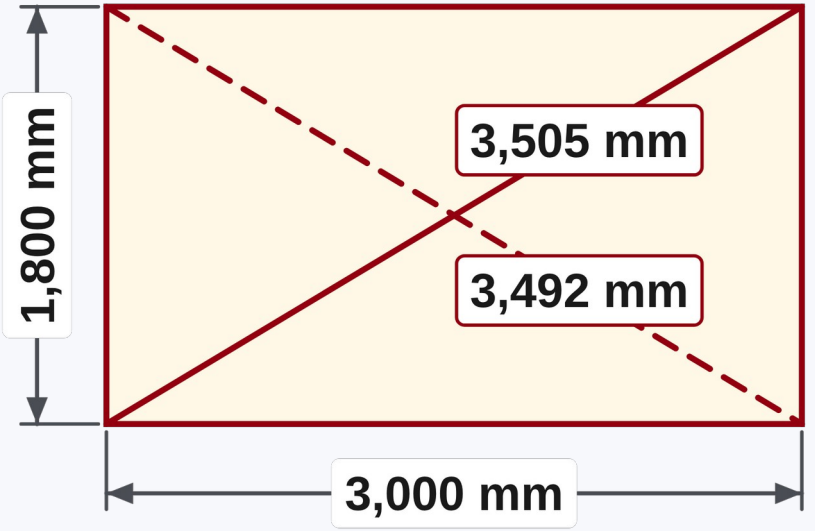
**Square: both diagonals = target**

Textbook sections 12.5–12.6 pp.304–306

# Use a target triangle and target diagonal

Scaled 3–4–5 • 1.2 m, 1.6 m and 2.0 m.
$1.2^2 + 1.6^2 = 4 = 2^2$. With the two legs held at their target lengths,
a 1.998 m diagonal is shorter than target: the corner is closed.

**Rectangle example • 3,000 mm × 1,800 mm.**
**Target diagonal = $\sqrt{(3{,}000^2 + 1{,}800^2)} \approx 3{,}499$ mm.**
**For the book's 3,505 / 3,492 mm readings, bring the ends of the longer diagonal closer together, then recheck dimensions and both diagonals.**




**Target ≈ 3,499 mm**

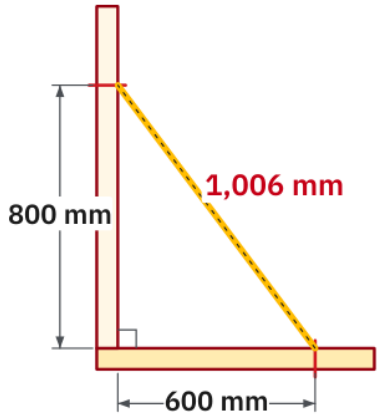
Together • Mark a paper or clear-floor triangle using the book dimensions.
Rotate measurer and checker roles. Use tape/chalk only—no cutting or power tools.

Exercise Book 12B Q4–5 pp.116–117 • Shape Canvas rectangle/diagonal and right triangle • Classroom layout activity

# Textbook | page 305

## Example 12.7 Is the frame square?

You mark 600 mm along one side of a frame corner and 800 mm along the other. The tape reads 1,006 mm between the marks. Is the corner square?




**SOLUTION**

1. For a square corner: $600^2 + 800^2 =$ $360{,}000 + 640{,}000 = 1{,}000{,}000$, and $\sqrt{1{,}000{,}000} = 1{,}000$ mm.
2. The tape reads 1,006 mm, which is 6 mm too long.
3. A diagonal that is too long means the corner is opened past 90°.

**Answer** **No. Close the corner until the marks are 1,000 mm apart.**

Textbook p.305 • Example 12.7 — Compare a measured diagonal with the square target

Textbook sections 12.5–12.6 pp.304–306

# Textbook | page 305

| Leg a | Leg b | Diagonal c | Check: $a^2 + b^2 = c^2$ |
|---|---|---|---|
| 300 mm | 400 mm | 500 mm | $90{,}000 + 160{,}000 = 250{,}000 = 500^2$ |
| 600 mm | 800 mm | 1,000 mm | $360{,}000 + 640{,}000 = 1{,}000{,}000 = 1{,}000^2$ |
| 3 ft | 4 ft | 5 ft | $9 + 16 = 25 = 5^2$ |
| 6 ft | 8 ft | 10 ft | $36 + 64 = 100 = 10^2$ |

**Table 12.1** Multiples of 3-4-5 in metric and imperial units.

Textbook p.305 • Table 12.1 — Scaled 3–4–5 checks

Textbook sections 12.5–12.6 pp.304–306

# Spot the mistake. Explain the correction.

## A triangle uses 3, 4 and 10 units. Is it a scaled 3–4–5 check?

Discuss the method before checking the answer.

**No. The same scale factor must apply to all three lengths, not just the diagonal.**

Textbook sections 12.5–12.6 pp.304–306

# Work together

In pairs, mark a paper or clear-floor triangle using the book dimensions. Rotate measurer/checker roles; no cutting or power tools.

**Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.**

Resources: Exercise Book 12A Q17–18 (p.115); 12B Q4–5,7 (pp.116–117)

Textbook sections 12.5–12.6 pp.304–306

# Your scheduled practice

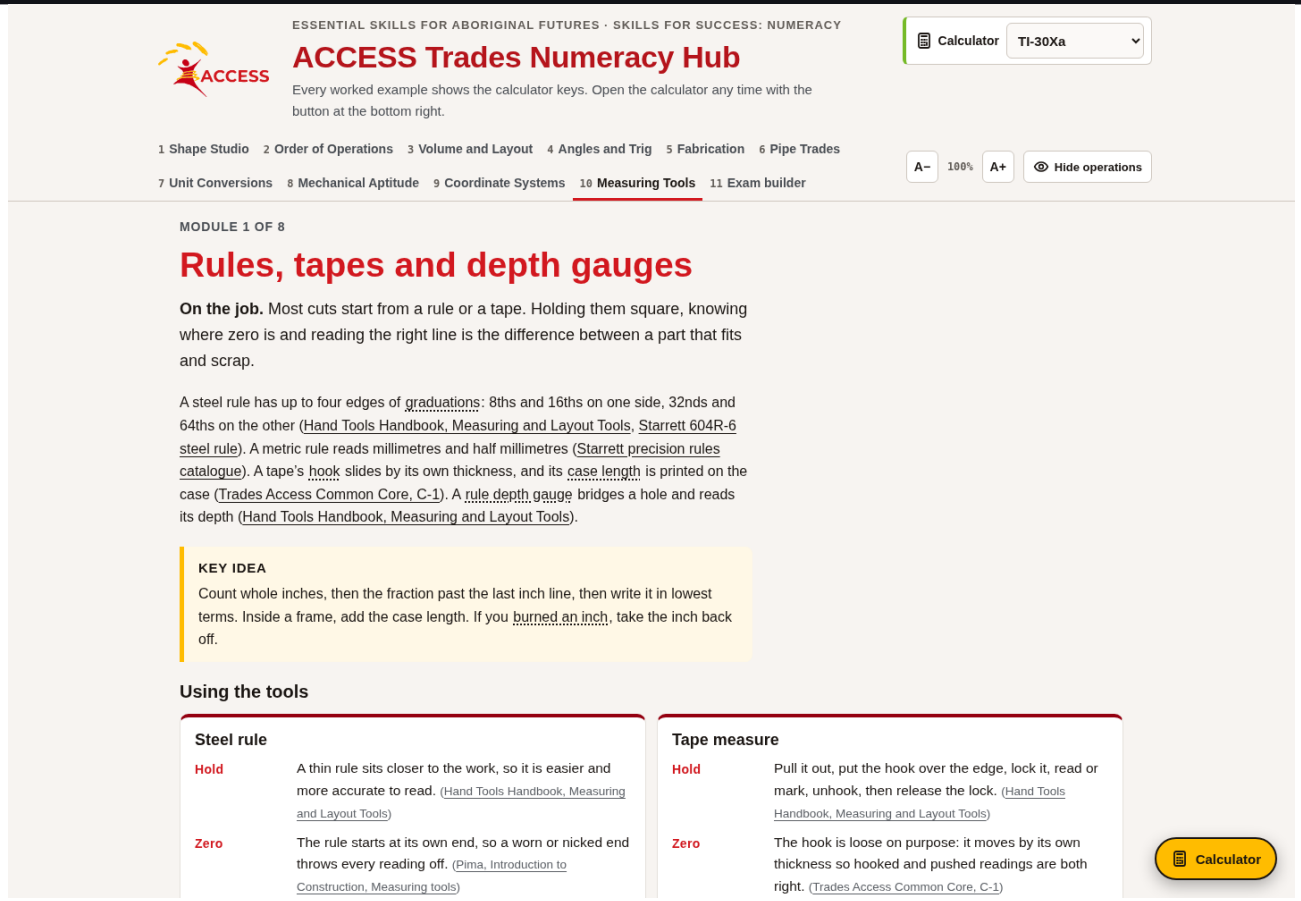
Solve Exercise Book 12B Q5,7 and 12A Q17–18. Compare theoretical diagonals to given readings; annotate the adjustment on a sketch.

**Solve on paper first; check with tools; correct one error and explain the change.**

Resources: Exercise Book 12A Q17–18 (p.115); 12B Q4–5,7 (pp.116–117)

Textbook sections 12.5–12.6 pp.304–306

# Open Numeracy Hub




Navigation example. Use the values and units in the assigned task.

Textbook sections 12.5–12.6 pp.304–306

# Practise in Numeracy Hub and Shape Canvas

1. Use a Shape Canvas right triangle for the scaled 3–4–5 check and a rectangle for diagonals.

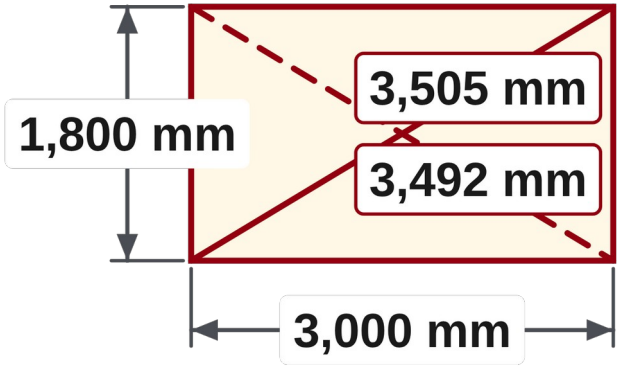
2. Mark the book dimensions on paper or a clear classroom floor with tape and chalk.

3. Compare target and measured lengths, using Numeracy Hub Measuring Tools square checks to review.

Textbook sections 12.5–12.6 pp.304–306

# Practise from the Exercise Book | 12B Q5

A pipe rack frame is 3,000 mm by 1,800 mm. The diagonals measure 3,505 mm and 3,492 mm. What should each diagonal measure, to the nearest millimetre? Which two corners do you push toward each other?




Each diagonal should be: ?

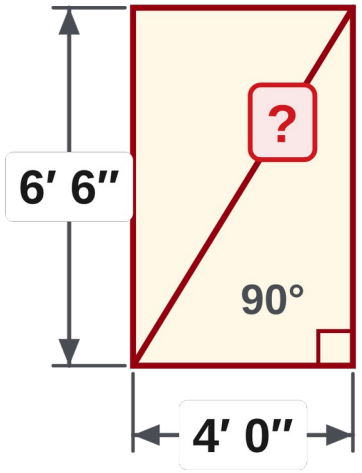
Corners to push together: ?

Write the operation, working and units before checking.

Textbook sections 12.5–12.6 pp.304–306

# Practise from the Exercise Book | 12B Q7

A gate frame is 4′ 0″ wide and 6′ 6″ tall. Find the length of the diagonal brace, corner to corner, to the nearest 1/16″. Give it in feet and inches.




Brace length: ?

Write the operation, working and units before checking.

Textbook sections 12.5–12.6 pp.304–306

# Check independently

Exercise Book 12B Q4 and Q5; state target lengths and the direction of correction.

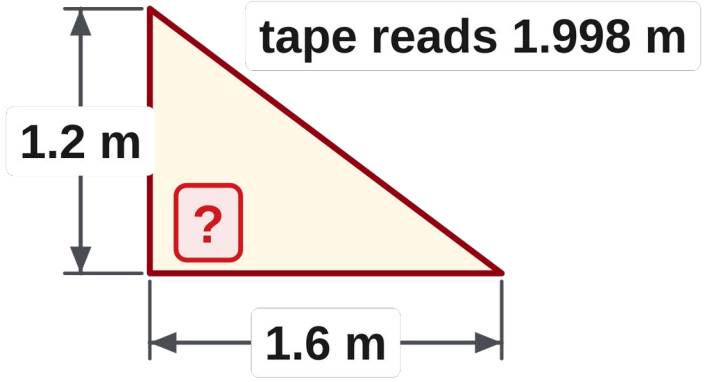
**Solve without displayed answers; include working, units and final rounding.**

Resources: Exercise Book 12A Q17–18 (p.115); 12B Q4–5,7 (pp.116–117)

Textbook sections 12.5–12.6 pp.304–306

# Independent check | 12B Q4

You square up a skid on the floor. You mark 1.2 m along one side and 1.6 m along the other. The tape reads 1.998 m between the marks. What should it read? Is the corner open or closed?




Tape should read: ?

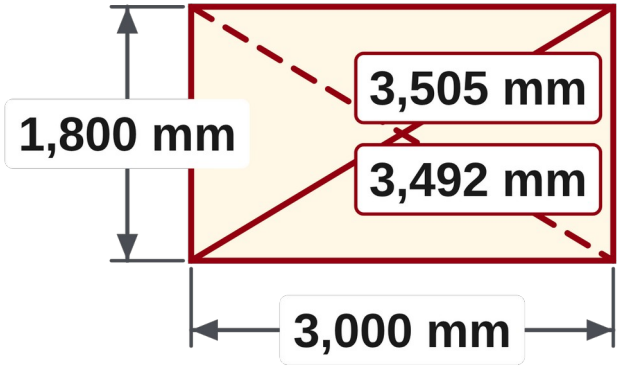
Corner open or closed: ?

Answer independently. Keep worked solutions closed.

Textbook sections 12.5–12.6 pp.304–306

# Independent check | 12B Q5

A pipe rack frame is 3,000 mm by 1,800 mm. The diagonals measure 3,505 mm and 3,492 mm. What should each diagonal measure, to the nearest millimetre? Which two corners do you push toward each other?




Each diagonal should be: ?

Corners to push together: ?

Answer independently. Keep worked solutions closed.

Textbook sections 12.5–12.6 pp.304–306

# Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

**Name the math and one other essential skill used; record a next practice step.**

Resources: Exercise Book 12A Q17–18 (p.115); 12B Q4–5,7 (pp.116–117)

Textbook sections 12.5–12.6 pp.304–306

# Break

## 2:00 PM–2:15 PM

Return ready for: SOH CAH TOA and missing sides

**This break is excluded from teaching minutes.**

# SOH CAH TOA and missing sides

**Today’s goal: Name sides relative to a marked angle and select sine, cosine or tangent to find a missing side.**

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

Exercise Book 13A Q1–8 (pp.127–128); 13C Q1–2 (p.131)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Recall and reconnect

Check hypotenuse identification; compare opposite and adjacent when the marked angle changes.

**Attempt the retrieval task independently; explain the method before checking.**

Resources: Exercise Book 13A Q1–8 (pp.127–128); 13C Q1–2 (p.131)

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

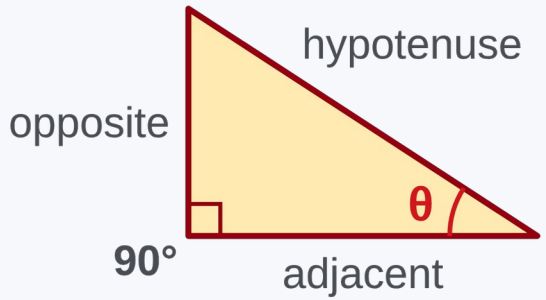
# Name sides from the chosen angle

**1** The hypotenuse is always opposite the right angle.

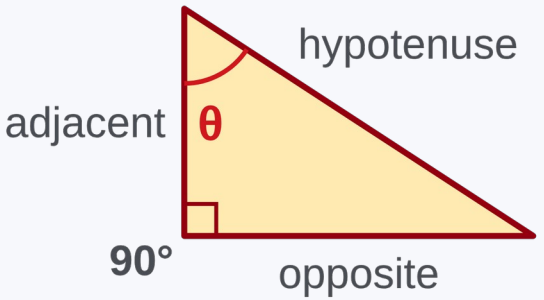
**2** The opposite side faces the marked angle.

**3** The adjacent side touches the marked angle and is not the hypotenuse.

**4** Changing the marked angle swaps opposite and adjacent.




**Angle θ marked at the bottom**




**Same triangle, θ at the top: names swap**

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

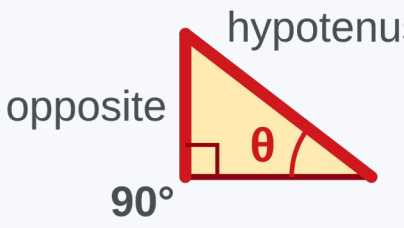
# Choose the ratio for the two sides

**1** SOH: sin θ = opposite/hypotenuse.

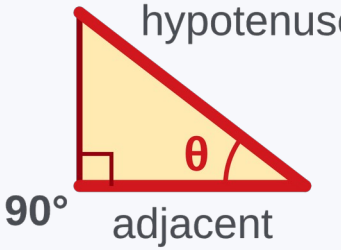
**2** CAH: cos θ = adjacent/hypotenuse.

**3** TOA: tan θ = opposite/adjacent.

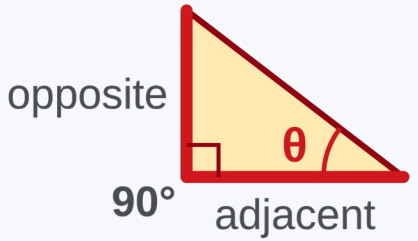
**4** Choose the known and wanted sides, rearrange, and use calculator DEG mode.




**SOH**
**sin θ =**
**opp ÷ hyp**




**CAH**
**cos θ =**
**adj ÷ hyp**

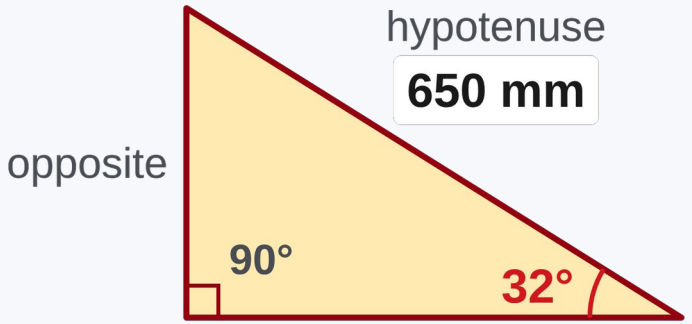



**TOA**
**tan θ =**
**opp ÷ adj**

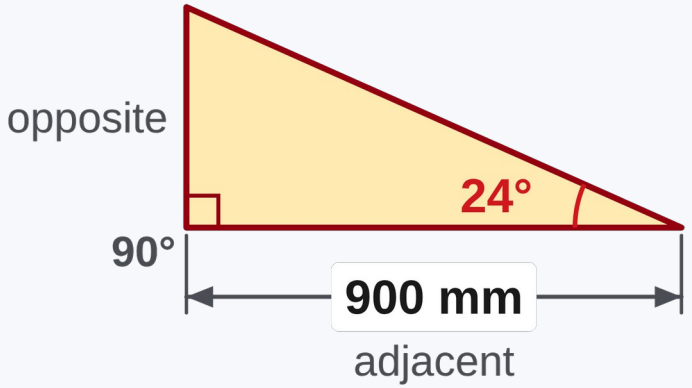
Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Label two sides before choosing the ratio

Known hypotenuse 650 mm; angle 32°; find the opposite side.
sin 32° = opposite ÷ 650 → opposite = 650 × sin 32° ≈ 344.45 mm.




**Known adjacent side 900 mm; angle 24°; find the opposite side.**
**tan 24° = opposite ÷ 900 → opposite = 900 × tan 24° ≈ 400.71 mm.**




Calculator check • Use DEG, not RAD. Follow the selected model's trig key order. Sketch the angle and side labels before entering keys.

Together • Work 13A Q1–5. Say the two relevant side names aloud. Choose the ratio, rearrange it, then calculate. Check the result against the sketch.

Exercise Book 13A pp.127–128 • Numeracy Hub → Angles and Trig → Module 3 • Shape Canvas right triangle

# Textbook | page 326




**Figure 13.1** The same triangle named from angle A (left) and from angle B (right). The opposite and adjacent sides swap. The hypotenuse stays the same.

Textbook p.326 • Figure 13.1 — Opposite and adjacent depend on the marked angle

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

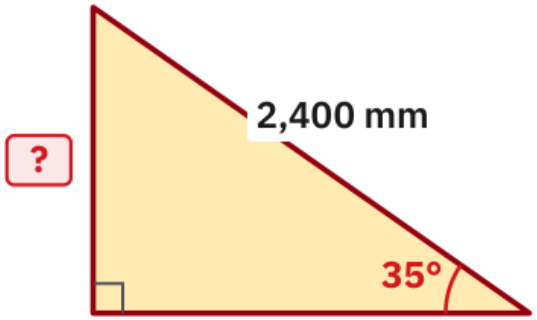
# Textbook | page 329

## Example 13.2 Height of a brace (sine)

A brace 2,400 mm long is set at 35° to the floor. How high up the wall does it reach? Give the answer to two decimal places.

**SOLUTION**

1. From the 35° angle: the brace is the hypotenuse, the height is the opposite side.
2. Opposite and hypotenuse: SOH. sin 35° = opposite ÷ 2,400.
3. Multiply both sides by 2,400: opposite = 2,400 × sin 35°.




Textbook p.329 • Example 13.2 — Sine and the height of a brace

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Spot the mistake. Explain the correction.

## Does the side drawn at the bottom always count as adjacent?

Discuss the method before checking the answer.

**No. Adjacent is defined relative to the marked angle; first identify the hypotenuse and that angle.**

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Work together

Work Exercise Book 13A Q1–5; say the two relevant sides before choosing a ratio.

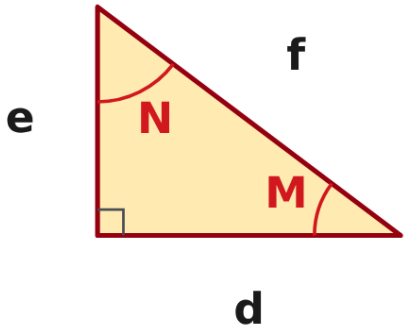
**Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.**

Resources: Exercise Book 13A Q1–8 (pp.127–128); 13C Q1–2 (p.131)

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Guided practice | 13A Q1

Look at the triangle. Which side is opposite angle M? Which side is adjacent to angle M? Which side is the hypotenuse? Which side is opposite angle N?

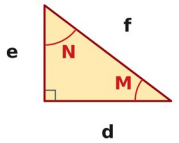



Work with a partner. Take turns explaining the next step.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Compare your method | 13A Q1

Look at the triangle. Which side is opposite angle M? Which side is adjacent to angle M? Which side is the hypotenuse? Which side is opposite angle N?




**CHECK TOGETHER**

**Opposite M: e. Adjacent to M: d. Hypotenuse: f. Opposite N: d.**

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Your scheduled practice

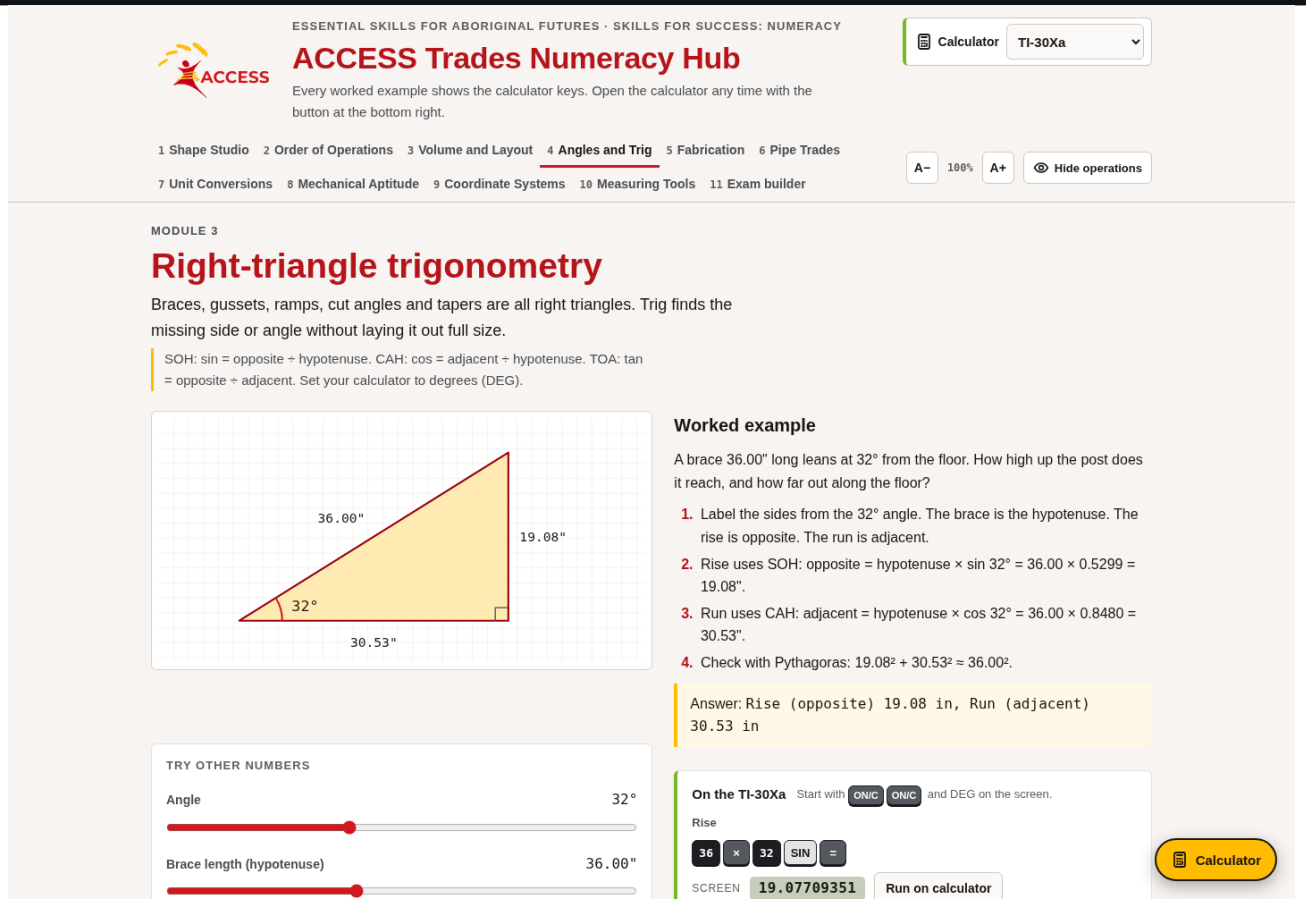
Complete Exercise Book 13A Q6–8 and 13C Q1–2 as ready. Use one Level 1 Numeracy Hub task and compare with the drawn triangle.

**Solve on paper first; check with tools; correct one error and explain the change.**

Resources: Exercise Book 13A Q1–8 (pp.127–128); 13C Q1–2 (p.131)

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Open Numeracy Hub




Navigation example. Use the values and units in the assigned task.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Practise in Numeracy Hub and Shape Canvas

1. Set the calculator to DEG and label the sides of a right triangle in Shape Canvas.

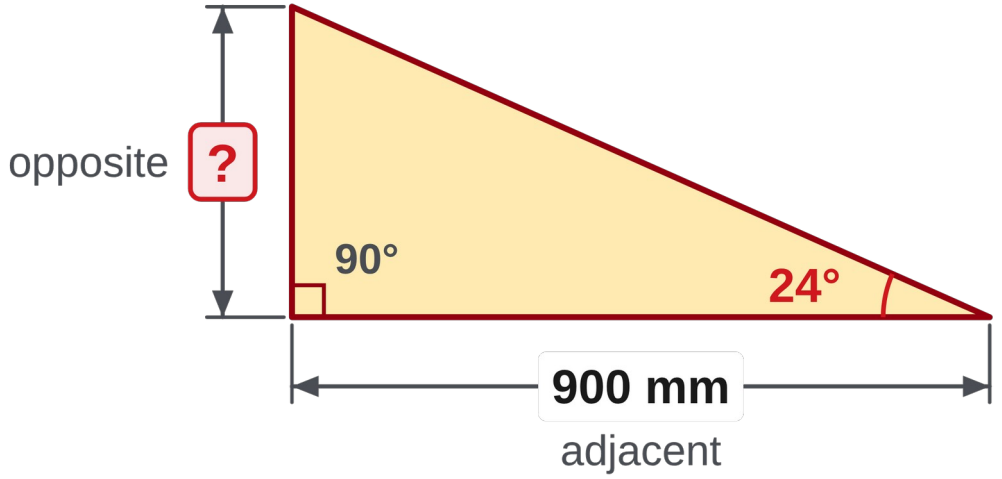
2. Hide one side; write the selected ratio and solve on paper.

3. Try the scheduled Level 1 Right-triangle trigonometry task in Numeracy Hub and compare diagrams.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Practise from the Exercise Book | 13A Q6

Adjacent 900 mm, angle 24°. Find the opposite side to two decimal places.

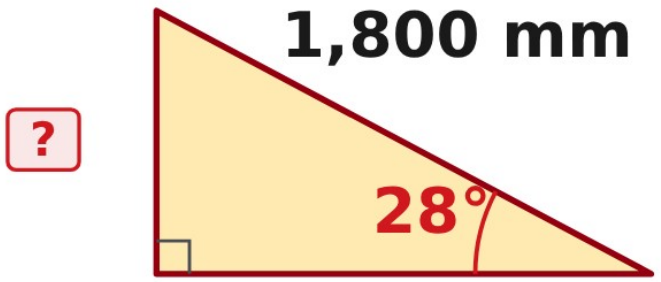



Write the operation, working and units before checking.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Practise from the Exercise Book | 13C Q1

Find the height to two decimal places.




Write the operation, working and units before checking.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Check independently

Exercise Book 13A Q4 and Q6; label the sides and chosen ratio.

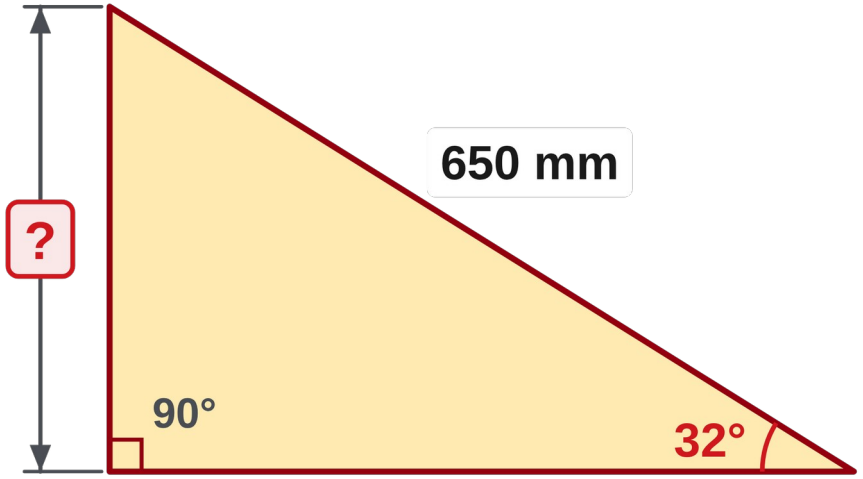
**Solve without displayed answers; include working, units and final rounding.**

Resources: Exercise Book 13A Q1–8 (pp.127–128); 13C Q1–2 (p.131)

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Independent check | 13A Q4

Hypotenuse 650 mm, angle 32°. Find the opposite side to two decimal places.

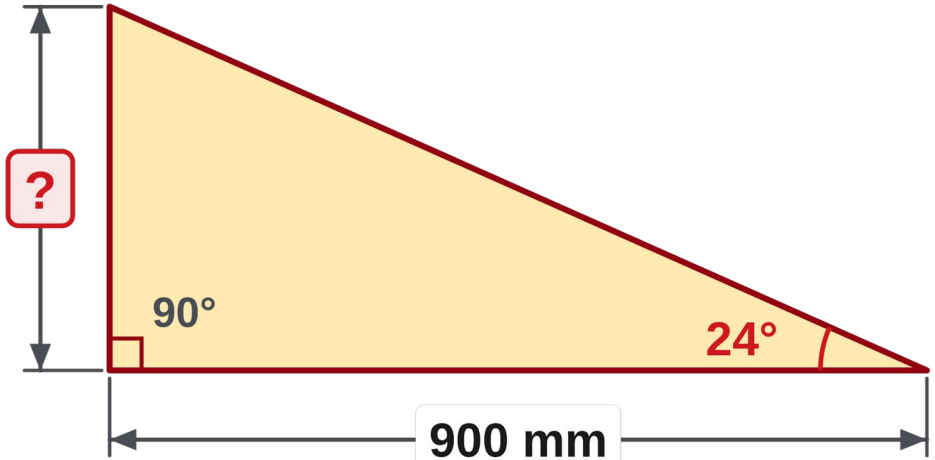



Answer independently. Keep worked solutions closed.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Independent check | 13A Q6

Adjacent 900 mm, angle 24°. Find the opposite side to two decimal places.




Answer independently. Keep worked solutions closed.

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58

# Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

**Name the math and one other essential skill used; record a next practice step.**

Resources: Exercise Book 13A Q1–8 (pp.127–128); 13C Q1–2 (p.131)

Textbook sections 13.1–13.3 pp.326–330; sections 2.10 pp.57–58